

STT 465 – Homework 2: Single-Parameter Bayesian Models

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Question 1: Bernoulli-Beta Posterior

[1 point]

A basketball player makes 18 of 25 free throws. Let $X_i \mid \omega \sim \text{Bern}(\omega)$, $i = 1, \dots, 25$, where ω is the player's underlying probability of making a free throw. Assume the prior $\omega \sim \text{Beta}(2, 2)$.

Throughout, let $n = 25$, $s = \sum_{i=1}^{25} x_i = 18$ (makes), and $n - s = 7$ (misses). The prior hyperparameters are $a = b = 2$.

- (a) Write the likelihood function $L(\omega \mid x)$ up to a multiplicative constant that does not depend on ω . [0.2 pt]

Since the X_i are conditionally independent given ω ,

$$L(\omega \mid x) = \prod_{i=1}^{25} \omega^{x_i} (1 - \omega)^{1-x_i} = \omega^{\sum x_i} (1 - \omega)^{n - \sum x_i} = \omega^{18} (1 - \omega)^7, \quad 0 \leq \omega \leq 1.$$

- (b) Use Bayes' theorem to derive the posterior distribution $\pi(\omega \mid x)$. Identify the posterior distribution and its parameters. [0.2 pt]

The $\text{Beta}(2, 2)$ prior density is

$$\pi(\omega) = \frac{1}{B(2, 2)} \omega^{2-1} (1 - \omega)^{2-1} \propto \omega(1 - \omega).$$

By Bayes' theorem,

$$\pi(\omega \mid x) = \frac{L(\omega \mid x) \pi(\omega)}{\int_0^1 L(\omega \mid x) \pi(\omega) d\omega} \propto \omega^{18} (1 - \omega)^7 \cdot \omega(1 - \omega) = \omega^{19} (1 - \omega)^8 = \omega^{20-1} (1 - \omega)^{9-1}.$$

This is the kernel of a Beta density, so after normalizing,

$$\boxed{\omega \mid x \sim \text{Beta}(20, 9)}, \quad \pi(\omega \mid x) = \frac{1}{B(20, 9)} \omega^{19} (1 - \omega)^8, \quad 0 < \omega < 1.$$

In general, a $\text{Beta}(a, b)$ prior with s successes in n trials gives a $\text{Beta}(a + s, b + n - s)$ posterior; here $(2 + 18, 2 + 7) = (20, 9)$.

- (c) Derive the posterior mean $E[\omega \mid x]$ and calculate its numerical value. [0.2 pt]

For $\omega \sim \text{Beta}(\alpha, \beta)$,

$$E[\omega] = \int_0^1 \omega \cdot \frac{\omega^{\alpha-1}(1-\omega)^{\beta-1}}{B(\alpha, \beta)} d\omega = \frac{B(\alpha+1, \beta)}{B(\alpha, \beta)} = \frac{\Gamma(\alpha+1)\Gamma(\alpha+\beta)}{\Gamma(\alpha)\Gamma(\alpha+\beta+1)} = \frac{\alpha}{\alpha+\beta},$$

using $\Gamma(z+1) = z\Gamma(z)$. With $\alpha = 20$, $\beta = 9$,

$$E[\omega | x] = \frac{a+s}{a+b+n} = \frac{20}{29} \approx 0.6897.$$

This lies between the prior mean $2/4 = 0.5$ and the sample proportion $18/25 = 0.72$, closer to the data because $n = 25$ outweighs the prior's effective sample size $a+b = 4$.

- (d) Calculate the posterior mode and posterior median. You may use `qbeta()` for the median. [0.2 pt]

For $\alpha, \beta > 1$, the Beta mode is found by setting the derivative of the log-density to zero:

$$\frac{d}{d\omega} [(\alpha-1)\log\omega + (\beta-1)\log(1-\omega)] = \frac{\alpha-1}{\omega} - \frac{\beta-1}{1-\omega} = 0 \implies \omega_{\text{mode}} = \frac{\alpha-1}{\alpha+\beta-2}.$$

So

$$\text{Mode} = \frac{19}{27} \approx 0.7037.$$

The median has no closed form, so we use the Beta quantile function:

```
qbeta(0.5, shape1 = 20, shape2 = 9)
#> [1] 0.6940678
```

$$\text{Median} \approx 0.6941.$$

Mean (0.6897) < median (0.6941) < mode (0.7037), consistent with a slightly left-skewed posterior ($\alpha > \beta$).

- (e) Use R to calculate $P(\omega > 0.70 | x)$. Include the R command you use. [0.2 pt]

$$P(\omega > 0.70 | x) = 1 - F_{\omega|x}(0.70) = \int_{0.70}^1 \frac{\omega^{19}(1-\omega)^8}{B(20,9)} d\omega.$$

```
1 - pbeta(0.70, shape1 = 20, shape2 = 9)
#> [1] 0.4724805

# Equivalent:
pbeta(0.70, shape1 = 20, shape2 = 9, lower.tail = FALSE)
```

$$P(\omega > 0.70 | x) \approx 0.4725.$$

Given the data and prior, there is about a 47% posterior probability that the player's true free-throw percentage exceeds 70%.

Question 2: Prior Sensitivity and Bayesian Updating [1 point]

Suppose a treatment is effective for 18 of 21 patients. Let ω denote the probability that the treatment is effective. Consider the two priors

$$\text{Prior A: } \omega \sim \text{Beta}(1, 1), \quad \text{Prior B: } \omega \sim \text{Beta}(4, 0.9).$$

18/21 successes, ω = probability of the treatment being effective. Posterior $\propto$ likelihood $\times$ prior.

(a) Derive the posterior distribution under Prior A. [0.2 pt]

The Beta prior has the form $\omega^{a-1}(1-\omega)^{b-1}$, so

$$\pi_A(\omega | x) \propto \omega^{18}(1-\omega)^3 \omega^{1-1}(1-\omega)^{1-1} = \omega^{19-1}(1-\omega)^{4-1}$$

$$\boxed{\omega | x \sim \text{Beta}(19, 4)}$$

(b) Derive the posterior distribution under Prior B. [0.2 pt]

Here the prior is $\omega^3(1-\omega)^{-0.1}$, so

$$\begin{aligned} \pi_B(\omega | x) &\propto \omega^{18}(1-\omega)^3 \omega^{4-1}(1-\omega)^{0.9-1} \\ &= \omega^{21}(1-\omega)^{2.9} \\ &= \omega^{22-1}(1-\omega)^{3.9-1} \end{aligned}$$

$$\boxed{\omega | x \sim \text{Beta}(22, 3.9)}$$

(c) For a general prior $\omega \sim \text{Beta}(a, b)$, show that the posterior mean can be written as

$$E[\omega | x] = \frac{a+b}{a+b+n} \cdot \frac{a}{a+b} + \frac{n}{a+b+n} \cdot \frac{x}{n}.$$

Explain what the two terms represent. [0.2 pt]

$\omega | x \sim \text{Beta}(a+x, b+n-x)$, and a $\text{Beta}(\alpha, \beta)$ has mean $\frac{\alpha}{\alpha+\beta}$. So

$$\begin{aligned} E[\omega | x] &= \frac{a+x}{(a+x)+(b+n-x)} = \frac{a+x}{a+b+n} = \frac{a}{a+b+n} + \frac{x}{a+b+n} \\ &= \boxed{\frac{a+b}{a+b+n} \cdot \frac{a}{a+b} + \frac{n}{a+b+n} \cdot \frac{x}{n}} \end{aligned}$$

$\frac{a}{a+b}$ is the prior mean, weighted by $\frac{a+b}{a+b+n}$. $\frac{x}{n}$ is the observed success proportion, weighted by $\frac{n}{a+b+n}$.

(d) Compute the posterior mean under Priors A and B. Briefly explain why the two answers differ. [0.2 pt]

$$E[\omega \mid x] = \frac{\alpha}{\alpha + \beta}.$$

Prior A, Beta(19, 4):

$$E[\omega \mid x] = \frac{19}{19 + 4} = \frac{19}{23} \approx \boxed{0.8261}$$

Prior B, Beta(22, 3.9):

$$E[\omega \mid x] = \frac{22}{25.9} \approx \boxed{0.8494}$$

They differ because the priors express different beliefs. Prior A has a lower mean, which means Prior A will pull the posterior mean down, below the observed success proportion.

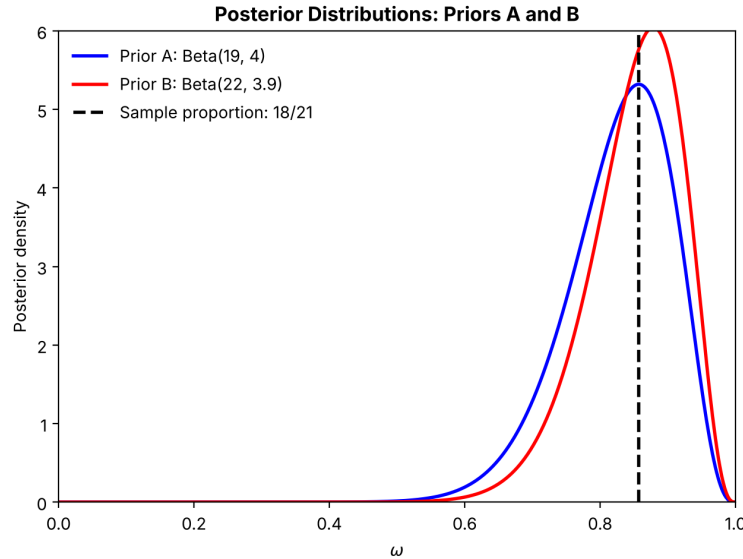
- (e) Use R to plot the two posterior densities on the same graph. Add a vertical line at the sample proportion $\tilde{\omega} = 18/21$. Briefly describe what the plot shows. [0.2 pt]

```
# Plot posterior densities
curve(dbeta(x, shape1 = 19, shape2 = 4),
      from = 0, to = 1,
      col = "blue", lwd = 2, ylim = c(0, 6),
      xlab = expression(omega),
      ylab = "Posterior density",
      main = "Posterior Distributions: Priors A and B")

curve(dbeta(x, shape1 = 22, shape2 = 3.9),
      add = TRUE, col = "red", lwd = 2)

# Mark the observed success proportion
abline(v = 18 / 21, col = "black", lty = 2, lwd = 2)

legend("topleft",
      legend = c("Prior A: Beta(19, 4)",
                  "Prior B: Beta(22, 3.9)",
                  "Sample proportion: 18/21"),
      col = c("blue", "red", "black"),
      lty = c(1, 1, 2), lwd = 2, bty = "n")
```



Both posteriors are concentrated near the sample proportion $18/21 \approx 0.857$ and overlap heavily. The Prior B posterior is shifted slightly to the right and is a little more peaked, while the Prior A posterior sits slightly lower. With 21 observations, the data dominate and the choice of prior has only a small effect.

Question 3: Credible Intervals and HPDI

[1 point]

Suppose that after observing data, the posterior distribution is $\omega \mid x \sim \text{Beta}(22, 4)$.

- (a) Write the R commands needed to calculate the 95% equal-tailed credible interval

$$[F_{\omega|x}^{-1}(0.025), F_{\omega|x}^{-1}(0.975)].$$

Report the numerical interval. [0.2 pt]

```
# 95% equal-tailed credible interval
equal_tailed <- qbeta(c(0.025, 0.975), shape1 = 22, shape2 = 4)
print(equal_tailed)
```

(0.68781, 0.95462)

- (b) Explain in one or two sentences the Bayesian interpretation of this 95% credible interval. [0.2 pt]

Given the observed data and the assumed model and prior, the posterior probability that ω lies between 0.68781 and 0.95462 is 95%.

- (c) Generate 10,000 posterior samples using `set.seed(1)` and `omega_post <- rbeta(10000, shape1 = 22, shape2 = 4)`. Use the `HDInterval` package to estimate the 95% HPDI. [0.2 pt]

```
# Install HDInterval automatically if it isn't already installed
if (!requireNamespace("HDInterval", quietly = TRUE)) {
  install.packages("HDInterval", repos = "https://cloud.r-project.org")
}

# 95% highest posterior density interval (HPDI)
set.seed(1)
omega_post <- rbeta(10000, shape1 = 22, shape2 = 4)

hpdi <- HDInterval::hdi(omega_post, credMass = 0.95)
print(hpdi)
```

[0.70898, 0.96719]

- (d) Compare the equal-tailed credible interval and the HPDI. Which one is shorter? [0.2 pt]

The shorter one is the HPDI, which has a smaller width.

- (e) Explain why the two intervals need not be identical for a skewed posterior distribution. [0.2 pt]

Beta(22, 4) is left-skewed, with a longer tail toward smaller values. Equal-tailed intervals put equal probability in both tails, but the HPDI prioritizes the highest-density values, so the HPDI shifts toward larger values.

Question 4: Comparing Two Population Proportions [1 point]

A study of births reports that among 980 births in one group, 437 were female. In a second group, among 820 births, 398 were female. Let ω_x = female birth proportion in group 1 and ω_y = female birth proportion in group 2. Assume independent priors $\omega_x \sim \text{Beta}(1, 1)$, $\omega_y \sim \text{Beta}(1, 1)$.

Group 1: 437/980 (ω_x); group 2: 398/820 (ω_y). In general, a Beta(a, b) prior gives a Beta($a + k$, $b + n - k$) posterior.

- (a) Derive the posterior distributions of ω_x and ω_y . [0.2 pt]

$$L(\omega_x | x) = \binom{980}{437} \omega_x^{437} (1 - \omega_x)^{543} \propto \omega_x^{437} (1 - \omega_x)^{543}$$

$$\pi(\omega_x) \propto \omega_x^{1-1} (1 - \omega_x)^{1-1} = 1$$

$$\pi(\omega_x | x) \propto \omega_x^{437} (1 - \omega_x)^{543}$$

$$\alpha - 1 = 437 \Rightarrow \alpha = 438, \quad \beta - 1 = 543 \Rightarrow \beta = 544$$

$$\boxed{\omega_x | x \sim \text{Beta}(438, 544)}$$

Similarly,

$$\pi(\omega_y | y) \propto \omega_y^{398} (1 - \omega_y)^{422} \omega_y^0 (1 - \omega_y)^0$$

$$\alpha - 1 = 398 \Rightarrow \alpha = 399, \quad \beta - 1 = 422 \Rightarrow \beta = 423$$

$$\boxed{\omega_y | y \sim \text{Beta}(399, 423)}$$

- (b) Calculate the posterior means $E[\omega_x | x]$ and $E[\omega_y | y]$. [0.2 pt]

$$E[\omega_x | x] = \frac{\alpha}{\alpha + \beta} = \frac{438}{438 + 544} \approx \boxed{0.4460}$$

$$E[\omega_y | y] = \frac{399}{822} \approx \boxed{0.4854}$$

- (c) Use the posterior simulation idea to estimate $P(\omega_x < \omega_y | x, y)$. Complete the missing parts of the R code. [0.2 pt]

```
set.seed(1)
U <- 10000

omega_x <- rbeta(
  U,
  shape1 = 438,
  shape2 = 544
)

omega_y <- rbeta(
  U,
  shape1 = 399,
  shape2 = 423
)

print(mean(omega_x < omega_y))
```

$$P(\omega_x < \omega_y | x, y) \approx 0.953$$

- (d) Explain in words what the probability $P(\omega_x < \omega_y | x, y)$ means. [0.2 pt]

It is the posterior probability that the female birth proportion for group 2 is greater than the female birth proportion for group 1, given x and y .

- (e) A frequentist analysis uses `prop.test(c(437, 398), c(980, 820), alternative = "less")`. Briefly explain the difference between the posterior probability in part (c) and the frequentist p -value. [0.2 pt]

```
# (e) A frequentist analysis uses
print(prop.test(
  c(437, 398),
  c(980, 820),
  alternative = "less"
))
```

The test gives $X^2 = 2.6371$, $df = 1$, and a p -value of 0.0522.

The difference is that the p -value only gives the probability, assuming the two population proportions are equal, of an observed difference at least as extreme as the one observed in the direction of group 1 having a lower proportion. Part (c), however, is the probability that group 1's true female birth rate is lower than group 2's, given the observed data and prior.

Question 5: Poisson-Gamma Model and Prediction [1 point]

Suppose the number of customer complaints received by a company on day i satisfies $X_i \mid \lambda \sim \text{Poi}(\lambda)$, $i = 1, \dots, n$. Over 20 days, a total of 54 complaints are observed: $n = 20$, $\sum_{i=1}^{20} x_i = 54$. Assume the Gamma prior $\lambda \sim \text{Ga}(a = 2, b = 1)$, where a is the shape and b is the rate.

$n = 20$, $\sum_{i=1}^{20} x_i = 54$, $\lambda \sim \text{Ga}(a = 2, b = 1)$. The posterior parameters are $a + \sum_i x_i$ and $b + n$.

- (a) Show that the likelihood function has the form $L(\lambda \mid x) \propto \lambda^{\sum_i x_i} \exp(-n\lambda)$. [0.2 pt]

$$L(\lambda \mid x) = \prod_i \frac{\lambda^{x_i} e^{-\lambda}}{x_i!} = \frac{\lambda^{\sum x_i} e^{-n\lambda}}{\prod_i x_i!}$$

$$L(\lambda \mid x) \propto \lambda^{54} e^{-20\lambda}$$

- (b) Derive the posterior distribution of λ and identify its parameters. [0.2 pt]

$$\pi(\lambda) \propto \lambda^{a-1} e^{-b\lambda} = \lambda^1 e^{-\lambda}$$

$$\pi(\lambda \mid x) \propto \lambda^{54+2-1} e^{-(20+1)\lambda}$$

$$\lambda \mid x \sim \text{Gamma}(56, 21)$$

- (c) Calculate the posterior mean $E[\lambda \mid x]$ and a 95% equal-tailed credible interval using `qgamma()`. [0.2 pt]

56 / 21

`qgamma(c(0.025, 0.975), shape = 56, rate = 21)`

$$E[\lambda \mid x] = \frac{56}{21} \approx 2.667$$

$$[2.01437, 3.40905]$$

- (d) Derive the posterior predictive mean for the number of complaints on a future day, $E[X_f \mid x]$. Explain why it is equal to $E[\lambda \mid x]$. [0.2 pt]

We know $X_f \mid \lambda \sim \text{Poisson}(\lambda)$, so $E[X_f \mid \lambda] = \lambda$.

$$E[X_f \mid x] = E_{\lambda \mid x}[E(X_f \mid \lambda, x)] = E_{\lambda \mid x}[\lambda]$$

$$E[X_f \mid x] = \frac{56}{21} \approx 2.667$$

Given λ , future counts follow a Poisson distribution with a mean of λ , and we average the conditional Poisson mean over the posterior distribution of λ .

- (e) Use posterior simulation to estimate $P(X_f \geq 4 \mid x)$. Report your estimate and explain what it means. [0.2 pt]

```
set.seed(1)
U <- 10000

lambda_post <- rgamma(
  U,
  shape = 56,
  rate = 21
)

x_future <- rpois(U, lambda = lambda_post)
mean(x_future >= 4)
```

$$\hat{P}(X_f \geq 4 \mid x) = 0.283$$

Given the observed data, there is approximately a 28.3% chance of receiving at least 4 complaints on a future day.